

ARQUITECTURA (2-2-1)									
ReLU									
LEARNING	CONVERGENCE			AVERAGE	AVERAGE	AVERAGE NON	AVERAGE	LEAST	
RATE	Converged	Not converged dead neurons	Not converged max epochs	COVERGENCE EPOCH	CONVERGENCE LOSS	CONVERGENCE LOSS	LOSS	LOSS	
0.005	28.80%	0.60%	70.60%	24217.20	0.050	1.505	1.086	0.0500	
0.015	29.00%	0.60%	70.40%	8071.76	0.050	1.510	1.086	0.0500	
0.044	29.00%	0.60%	70.40%	2751.86	0.050	1.510	1.086	0.0500	
0.130	29.00%	0.80%	70.20%	931.52	0.050	1.511	1.087	0.0499	
0.385	29.00%	0.80%	70.20%	314.68	0.050	1.511	1.087	0.0498	
1.140	28.80%	0.80%	70.40%	106.28	0.050	1.512	1.091	0.0494	
3.376	25.80%	9.00%	65.20%	35.98	0.049	1.627	1.220	0.0479	
10.000	2.20%	75.80%	22.00%	11.64	0.047	5.602	5.480	0.0448	

Leaky ReLU								
LEARNING	CONVERGENCE	SATURATION	AVERAGE	AVERAGE	AVERAGE NON	AVERAGE	LEAST	
RATE	RATE	RATE	COVERGENCE EPOCH	CONVERGENCE LOSS	CONVERGENCE LOSS	LOSS	LOSS	
0.005	33.00%	0.60%	41825.10	0.050	1.875	1.273	0.0500	
0.015	33.20%	0.40%	13905.70	0.050	1.881	1.273	0.0500	
0.044	33.00%	1.40%	4496.46	0.050	1.875	1.273	0.0500	
0.130	33.00%	0.60%	1524.30	0.050	1.882	1.277	0.0499	
0.385	33.20%	0.40%	544.13	0.050	1.882	1.274	0.0498	
1.140	32.20%	0.40%	148.19	0.050	1.887	1.296	0.0495	
3.376	37.00%	2.60%	96.04	0.049	1.897	1.213	0.0484	
10.000	11.60%	83.60%	81.19	0.024	23.380	20.671	0.0000	

Standard Sigmoid								
LEARNING	CONVERGENCE	SATURATION	AVERAGE	AVERAGE	AVERAGE NON	AVERAGE	LEAST	
RATE	RATE	RATE	COVERGENCE EPOCH	CONVERGENCE LOSS	CONVERGENCE LOSS	LOSS	LOSS	
0.005	71.40%	74.20%	198564.00	0.050	1.394	0.434	0.0500	
0.015	71.60%	74.20%	66267.90	0.050	1.403	0.434	0.0500	
0.044	72.00%	74.20%	22641.30	0.050	1.413	0.432	0.0500	
0.130	71.80%	74.20%	7646.31	0.050	1.413	0.434	0.0500	
0.385	71.60%	74.20%	2577.57	0.050	1.413	0.437	0.0500	
1.140	71.40%	74.00%	871.77	0.050	1.413	0.440	0.0499	
3.376	71.80%	73.80%	295.81	0.050	1.417	0.435	0.0498	
10.000	49.60%	88.80%	109.26	0.050	1.470	0.765	0.0493	

Tanh							
LEARNING	CONVERGENCE	SATURATION	AVERAGE	AVERAGE	AVERAGE NON	AVERAGE	LEAST
RATE	RATE	RATE	COVERGENCE EPOCH	CONVERGENCE LOSS	CONVERGENCE LOSS	LOSS	LOSS
0.005	60.80%	1.20%	55262.10	0.050	1.405	0.581	0.0500
0.015	61.00%	1.20%	18418.50	0.050	1.412	0.581	0.0500
0.044	61.00%	1.20%	6279.50	0.050	1.412	0.581	0.0500
0.130	61.00%	1.20%	2125.94	0.050	1.412	0.581	0.0500
0.385	61.00%	1.20%	718.60	0.050	1.412	0.581	0.0499
1.140	61.60%	1.00%	243.96	0.050	1.413	0.573	0.0498
3.376	63.20%	3.20%	87.00	0.050	1.405	0.549	0.0493
10.000	24.20%	70.60%	43.30	0.048	4.106	3.124	0.0283

CONVERGENCE COMPARATION

Learning Rate	ReLU	Leaky ReLU	Standard Sigmoid	Tanh
0.005	28.80%	33.00%	71.40%	60.80%
0.015	29.00%	33.20%	71.60%	61.00%
0.044	29.00%	33.00%	72.00%	61.00%
0.130	29.00%	33.00%	71.80%	61.00%
0.385	29.00%	33.20%	71.60%	61.00%
1.140	28.80%	32.20%	71.40%	61.60%
3.376	25.80%	37.00%	71.80%	63.20%
10.000	2.20%	11.60%	49.60%	24.20%



